Thoughts on Different Research Styles

Looking back on my years in graduate school, I realize how much I learned from the different research styles and personalities of my professors across different areas of maths. From my algebra professors, I learned the importance of working carefully through details, understanding things, finding simple and elegant proofs, and developing a particular perspective on the subject. From my number theory professors, I learned to take a broad and open-minded approach to different ideas and methods, and, even more, to find and formulate questions, and to have the patience to start with small questions and gradually build them up.

In commutative algebra, one thing I have always appreciated about my algebra professors is seeing how in practice the simplest proofs can sometimes reflect the deepest understanding. A proof is not only a certificate that a statement is true; the way it is organized can reveal how one sees the underlying structures. This is part of what makes commutative algebra so beautiful to me. Its language can be remarkably simple, but there are hidden structures and rich mathematical phenomena beneath it. There is something deeply aesthetic about the subject to me. And I don't feel that the way I approach and enjoy maths will change very much despite the development of AI.

Personally, I prefer not to use AI when doing maths research, although I find it quite useful in many other aspects of life. When it comes to maths, the process feels different to me, perhaps because of my strong attachment to the intellectual traditions I inherited. More generally, when it comes to things I have a strong sense of craftsmanship about, I prefer not to involve AI in a substantial way. That said, I don't think collaborating with AI is a bad thing. When used responsibly, it can be a different research style that stimulates new ways of thinking. Whether and how mathematicians choose to use AI may depend on personal preferences and perhaps also on the particular area of maths one works in.

I also don't think that mathematicians who choose not to use AI will necessarily be at a disadvantage in their research or find that all their problems have already been solved by AI. There are always plenty of questions to ask and investigate beyond a stated conjecture, many of which have not even been formulated explicitly or made known to others. Sometimes seeing an interesting problem and exploring where it leads is itself the beginning of a new development. In this sense, I don't think there is any need to feel that one is racing against AI.

Mathematical Insights

Mathematical research involves more than proving difficult statements. I think what makes a conjecture interesting is often the possibility of exploring new mathematical territory, either along the way to a solution or through what the solution itself might reveal. Sometimes a proof reveals the underlying structure clearly, while other times it settles the conjecture without explaining nearly as much. A counterexample can leave even more questions behind, since it often tells us that something fails without explaining why or under what conditions the conjecture might still hold.

Sometimes a mathematician feels that a theorem has been formulated in the wrong way, or that a proof fails to explain the underlying phenomenon. I often heard my professors say, "There should be a more conceptual explanation," or "There should be a much simpler proof from the right perspective." Such judgments arise from a mathematical understanding and intuition developed through experience.

That internal picture develops slowly and individually. Over time, we develop a personal sense of

how a subject behaves, shaped by the problems we have worked on, the mistakes we have made, and the different perspectives we have encountered. Mathematical intuition involves recognizing what matters and sensing the underlying structures. There is something deeply personal about this process.

AI's ability to draw on a large amount of existing literature certainly has its advantages, but it does not necessarily lead to the most insightful approach. Of course, things can be quite different when mathematicians bring their own understanding and perspectives to AI-generated ideas, refining them or developing them in new directions.

Collaboration with AI

Mathematicians have always had different ways of working. Some prefer frequent discussion and collaboration, while others work mostly alone. Similarly, some may find constant interaction with AI productive, while others may prefer not to use it at all. These choices depend on individual preferences and perhaps also on the areas of mathematics in which one works. I think mathematicians should have the freedom to develop their own research styles.

AI also introduces new forms of mathematical collaboration. Mathematicians can genuinely engage with AI-generated ideas, bringing their own understanding and perspectives to refine and develop them. In such cases, it may be difficult to draw a sharp distinction between human and AI contributions, just as intellectual contributions in traditional collaborations are not always easily separated.

Of course, there may also be cases where AI makes the predominant contribution. I think this is a different situation, but such work can still be appreciated if the human authors have assimilated the ideas and developed a comparable understanding. Whatever research style one chooses, mathematical understanding remains essential.

Whatever forms the mathematical community may find to adapt to these changes, I think the values that make maths meaningful will remain, and may even be deepened as we reconsider the relationship between humans and AI. I hope that everyone can continue to enjoy maths just as much, regardless of how AI develops.

L. X.
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